

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Science case: Variable stars

Some of the earliest variables discovered: Mira Ceti

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- David Fabricius discovered “omicron ceti” in 1596:
 - It brightened from the 3rd to the 2nd magnitude, then disappeared. Actually, it disappeared from view since it faded beyond the detection limit.
 - It was re-observed in 1609 by Fabricius, and re-discovered in 1631 by Johann Fokkens Holwarda who determined a 11 month period.
 - Johannes Hevelius observed it in 1639 and 1642, and named the star “Mira” – The Wonderful.
- Earlier ancient discovery records unclear. Hipparchus may have discovered Mira in 134 BC.

Some of the earliest variables discovered: Algol

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- Geminiano Montanari discovered “beta Persei” in 1667:
 - Brightness varies from 2.1 to 3.4 magnitude in 2.87 days.
 - Period of less than 3 days credited to John Goodricke in 1782-83.
 - In 1881 Edward Pickering theorized (correctly) that it was an eclipsing binary star system.
- European, Arab, and Chinese cultures knew its variability and considered it evil:
 - “Demon’s Head” – Arab
 - “Mischief-maker” – Arab
 - “Satan’s Head” – Hebrew
 - “Lilith” – Adam’s legendary demonic first wife – Babylon/Hebrew
 - “The Spectre’s Head” - Europe
 - “Piled-up Corpses” - Chinese

Some of the earliest variables discovered: Supernovae

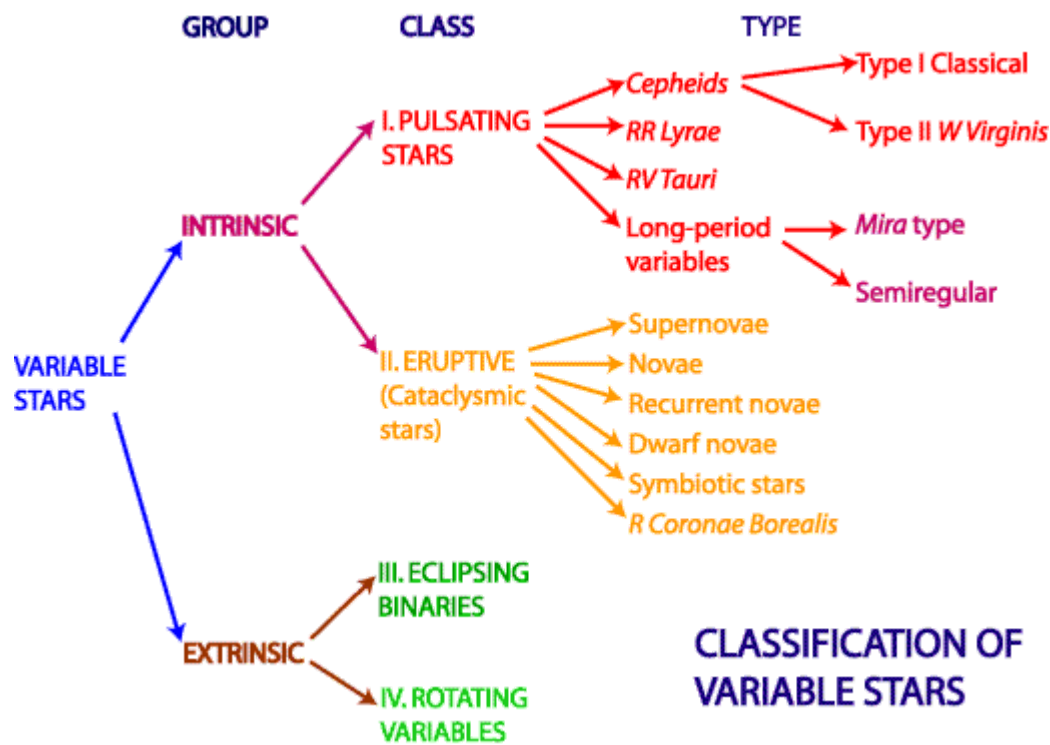
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- Chinese astronomers have been the first (maybe...) to record a supernova:
 - Over 20 historical candidates identified over the past 2000 years.
 - Confirmed dates include 185, 393, 1006 (brightest and also recorded in Egypt, Iraq, Italy, Japan and Switzerland), 1054 (Crab nebula remnant shown above).
- In 1572 Tycho Brahe observed SN 1572 and argued it was very far from Earth.

A rich zoo!

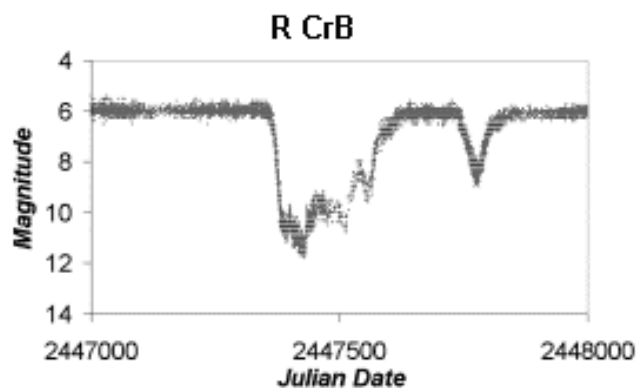
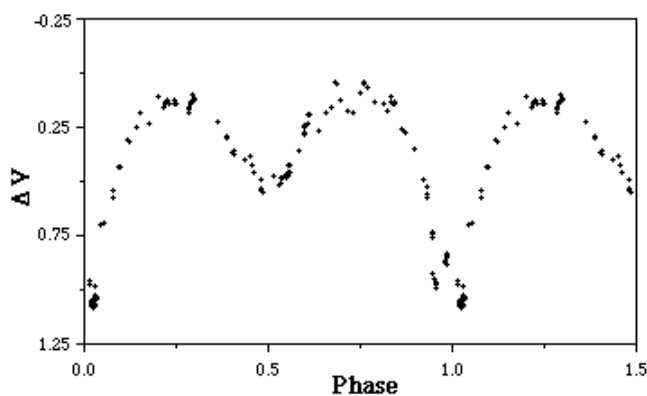
.....

- Pulsating Variables:
 - Periodic expansion and contraction of their structure (Cepheids, RR Lyrae, RV Tauri, Long Period, Semi-regular, etc.).
- Eclipsing Variables
- Eruptive Variables:
 - Supernovae, novae, dwarf novae, cataclysmic variables, etc. (we don't discuss them here).



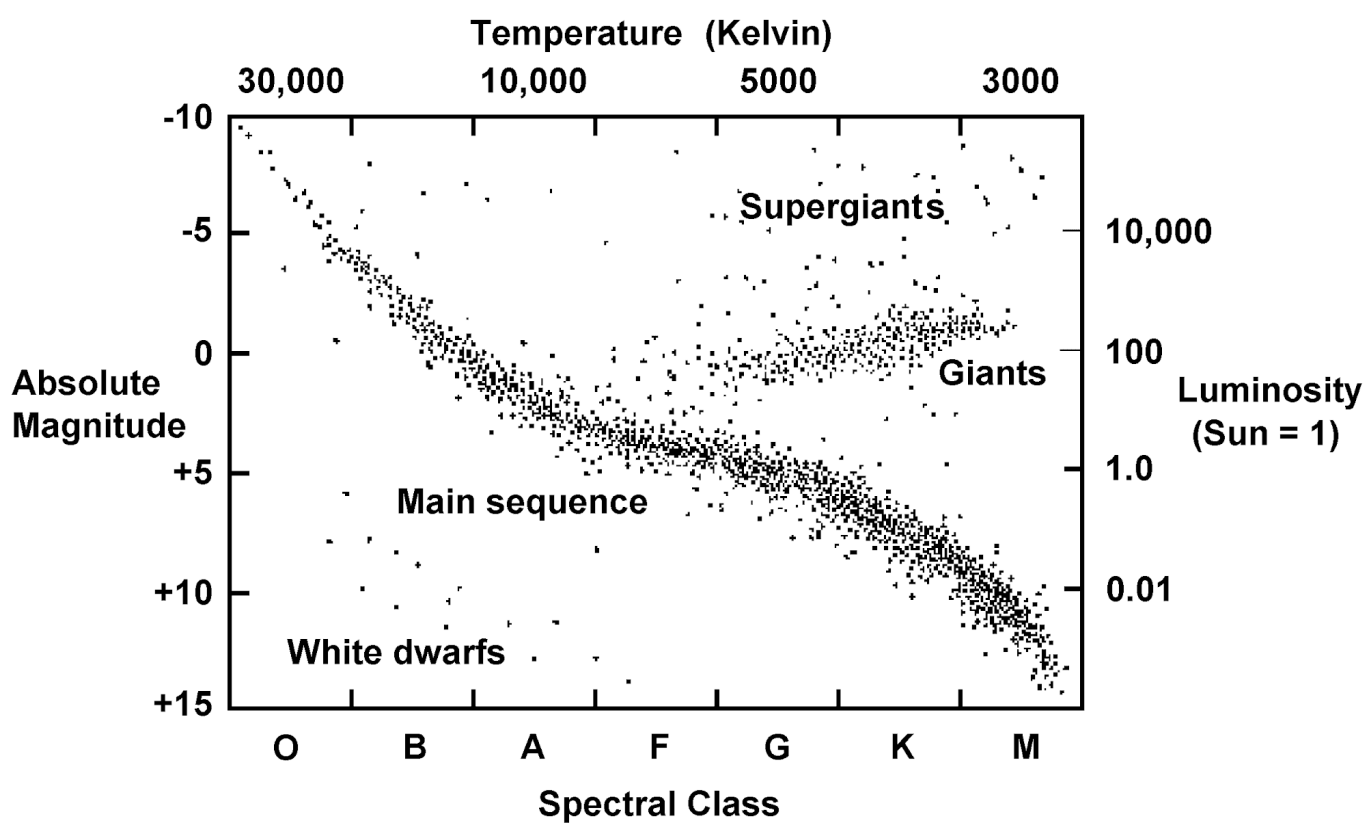
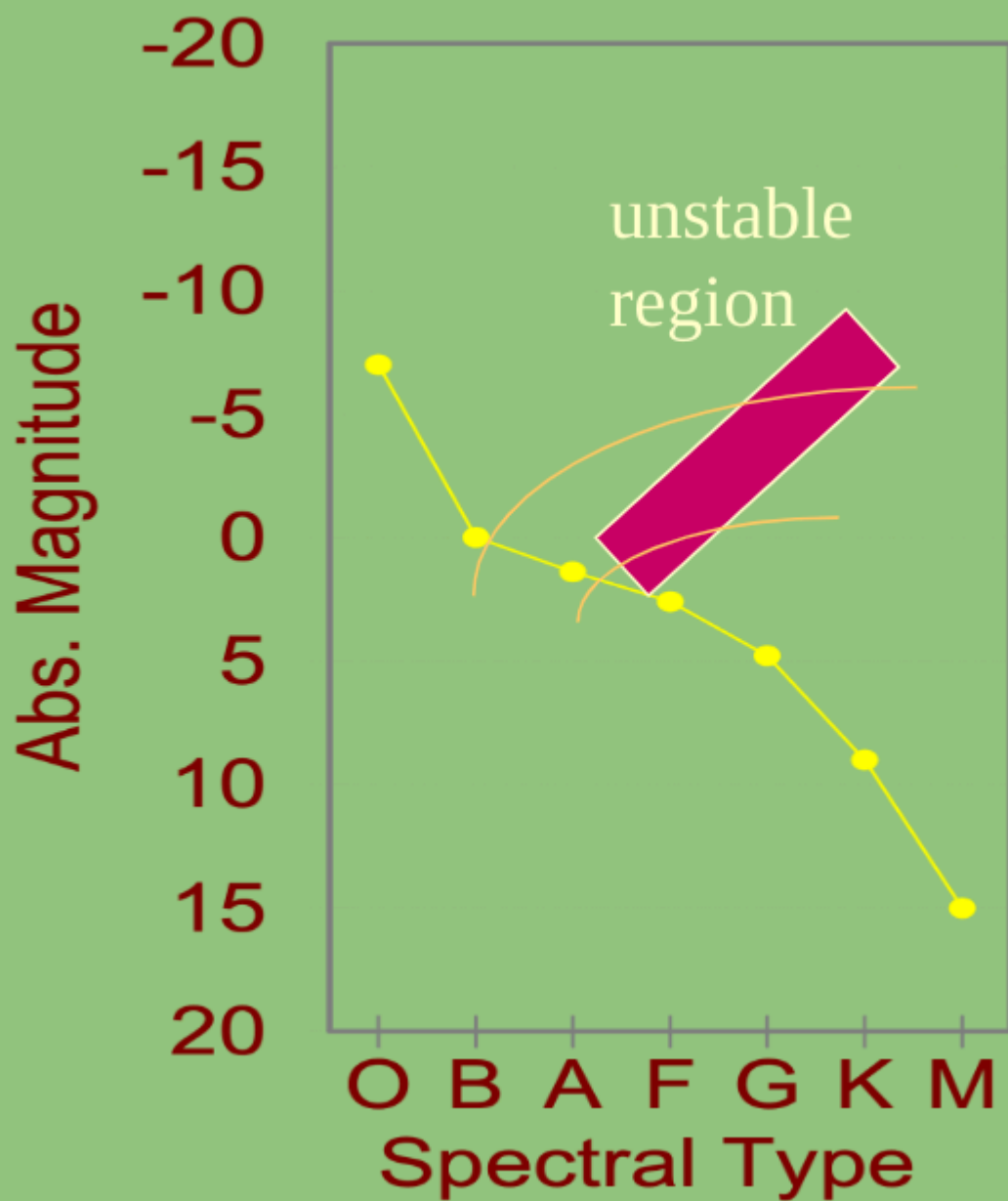
Variable star properties

- Light curves are used to track the luminosity of a variable star with time.
 - Measure period and amplitude
- Some stars are very regular, while others are irregular.



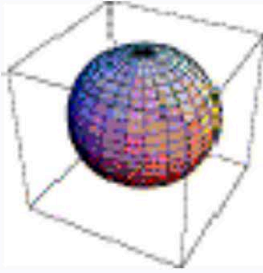
Intrinsic variables

- As giants, stars can become unstable. The temperature and luminosity fluctuate as the gas expands and contracts.
 - These are pulsating variables.

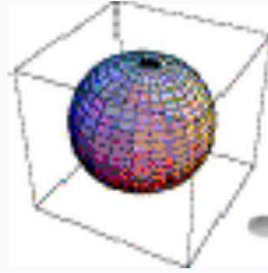


Pulsating variables: radial and non-radial modes

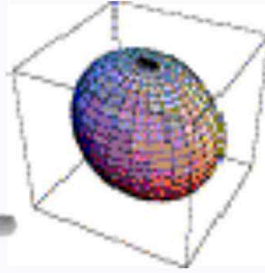
$l=1$ $m=0$



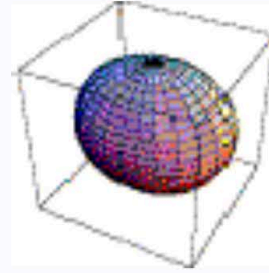
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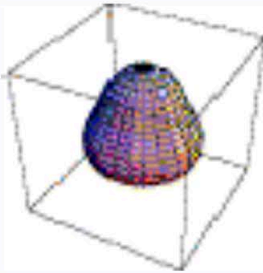
$l=2$ $m=1$



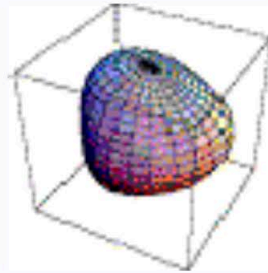
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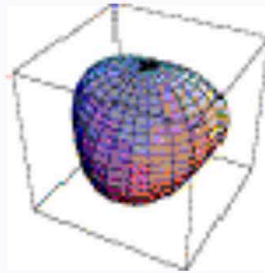
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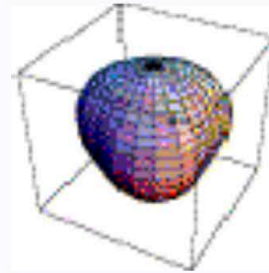
$l=3$ $m=1$



$l=3$ $m=2$

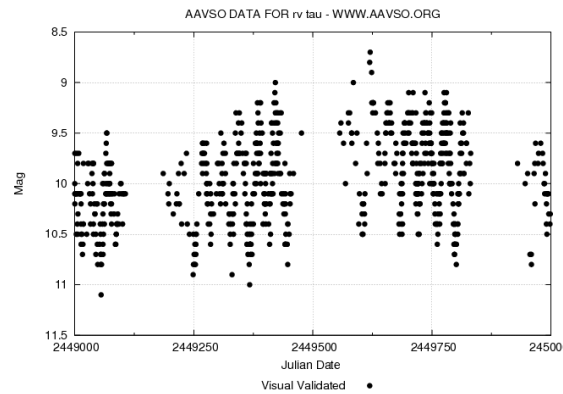
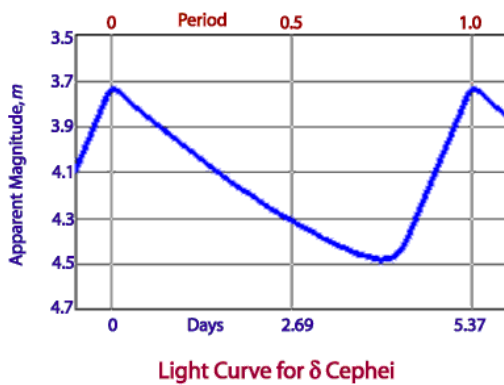


$l=3$ $m=3$



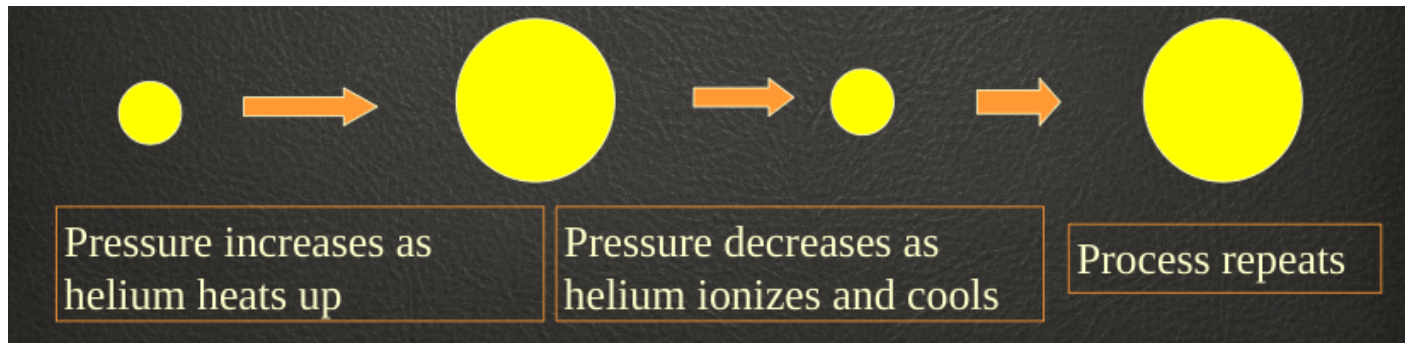
Examples of different types of pulsating variables

- Cepheids Period 1-70 day (on the left) and RV Tauri – Period up to 100 days (on the right)

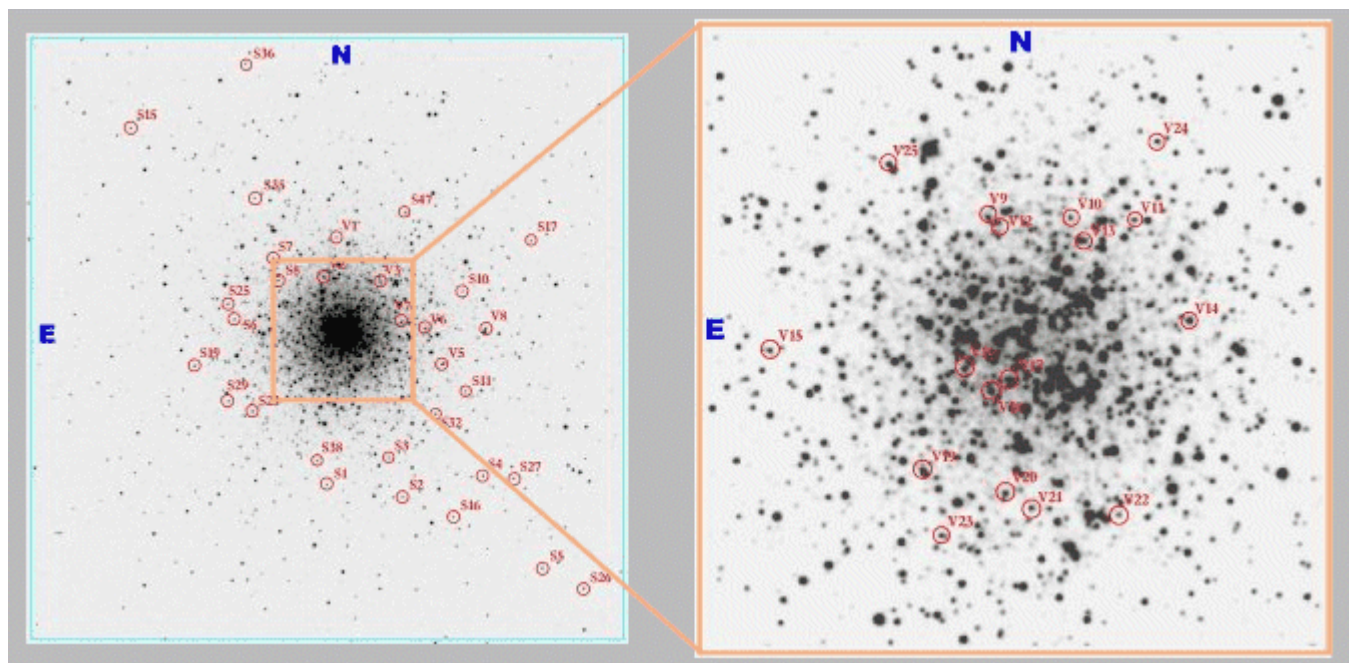


Variables as standard candles

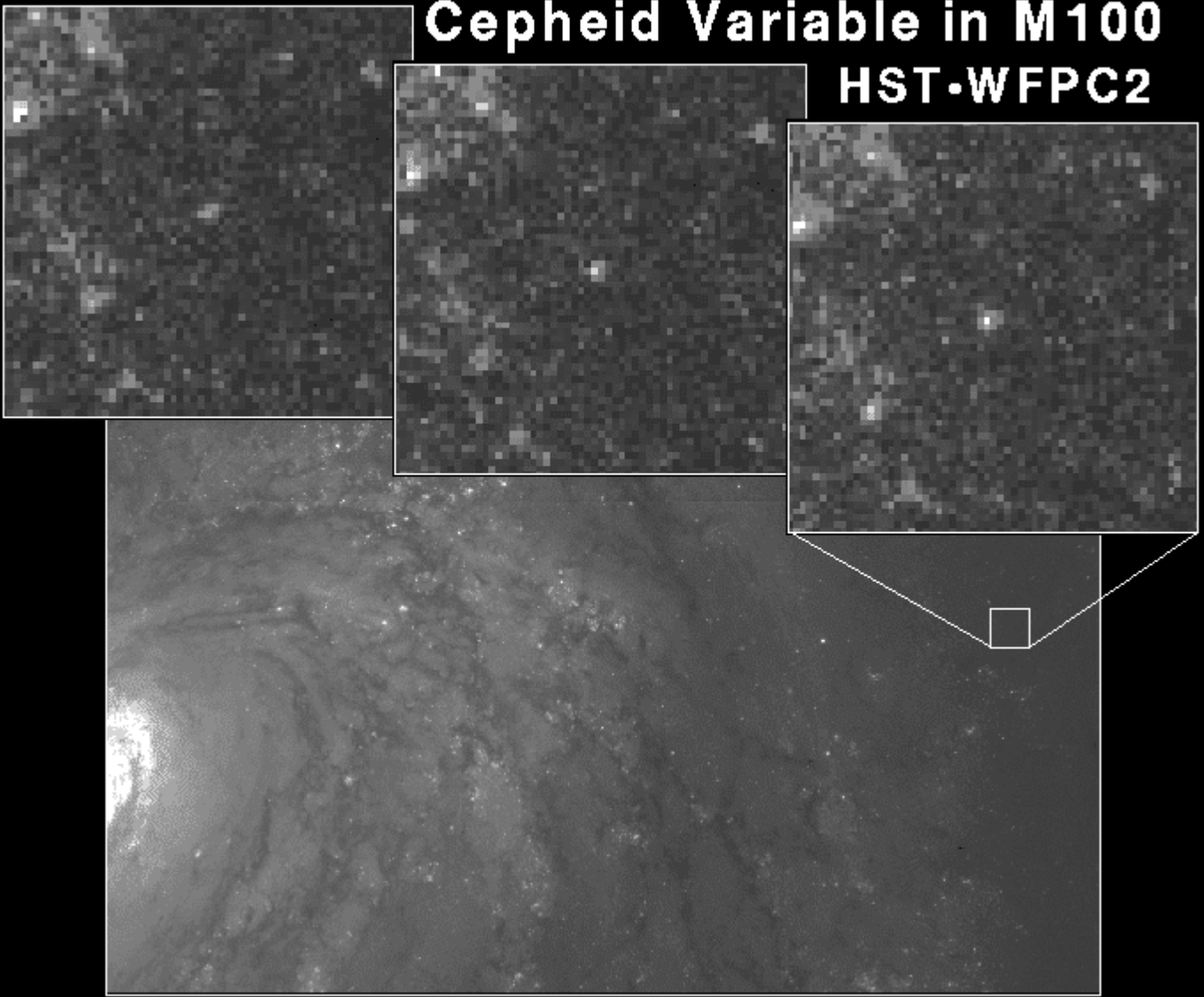
- The best measure of large distances are variable stars.
 - Pressure and radiation precisely oscillate and luminosity is directly related to the period.
 - The apparent magnitude and period give the distance.



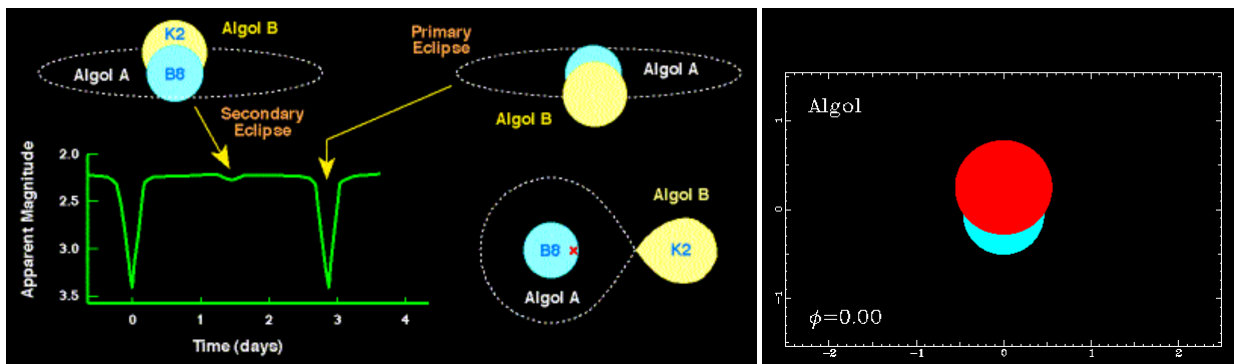
- *RR-Lyrae* stars are low mass pulsating variables. About $1M_{\odot}$ with period less than a day often found in clusters.
- Large mass stars that pass through the unstable region are known as *Cepheid variables*.
 - They have From $5M_{\odot}$ to $20M_{\odot}$ with period of 1 to 100 days.
 - Cepheids are important to measure distances to other galaxies.
- The pulsating mechanism in star's interior is due to He^+ loosing e^- producing He_2^+ . As soon the star expands, it becomes brighter and cooler, and $\text{He}_2^+ + e^-$ to give He^+ . And the cycle repeats.



Cepheid Variable in M100 HST-WFPC2

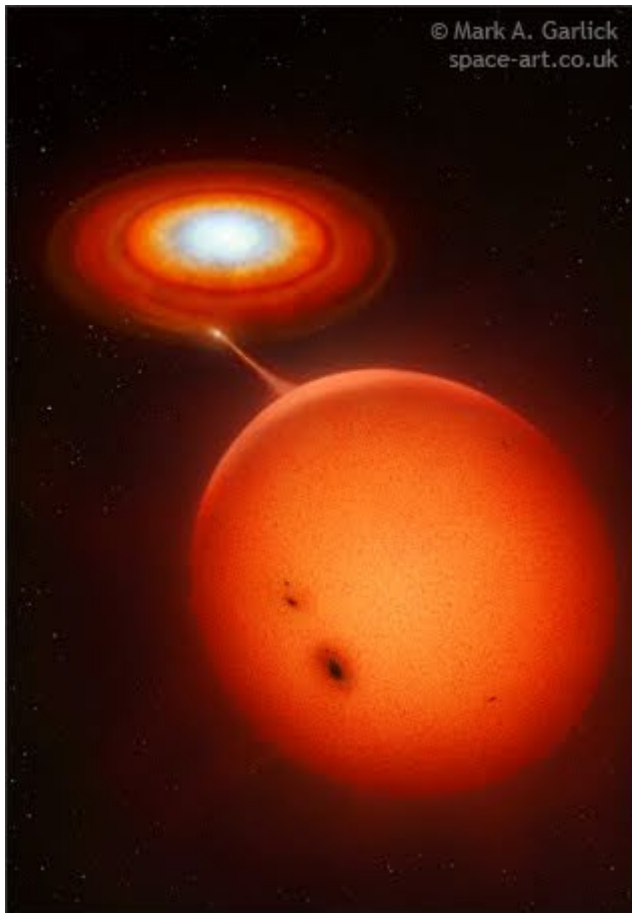


Eclipsing variables



Erupting variables

- Erupting variables are stars that have occasional violent outbursts caused by thermonuclear processes either in their surface layers or deep within their interiors.
- Major Types:
 - Supernovae
 - Recurrent Novae
 - Cataclysmic Variables
- Key Properties of CV:
 - Binary Star System.
 - Stellar Material Flows from red dwarf star onto accretion disk surrounding white dwarf companion.
 - Flow stops and starts.
 - Orbital period 78 minutes up to around 10 hours.
 - “Hot spot” where stream hits accretion disk is often hotter and brighter than either star.

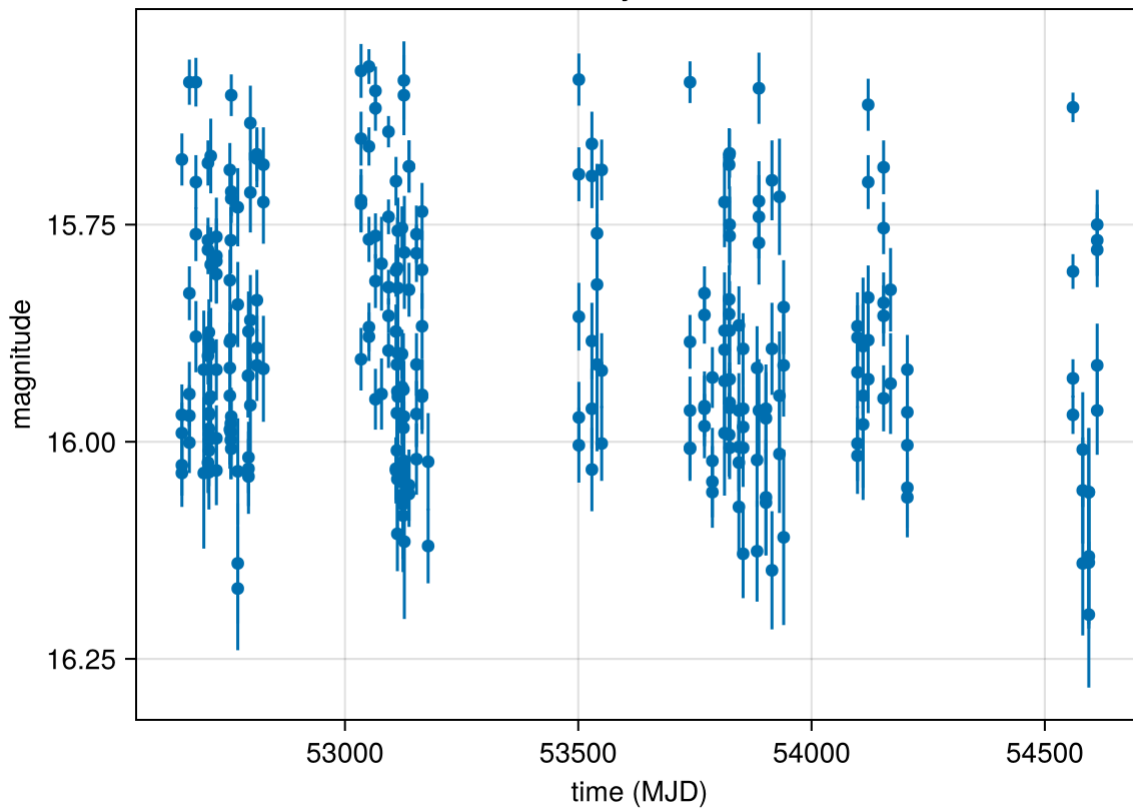


Let's study a variable source from the ground based LINEAR survey: 11375941

	t	mag	magerr
1	52650.4	15.969	0.035
2	52650.4	16.036	0.039
3	52650.5	15.99	0.035
4	52650.5	16.027	0.035
5	52650.5	15.675	0.03

Length of the monitoring: **5.37** years.

LINEAR object 11375941

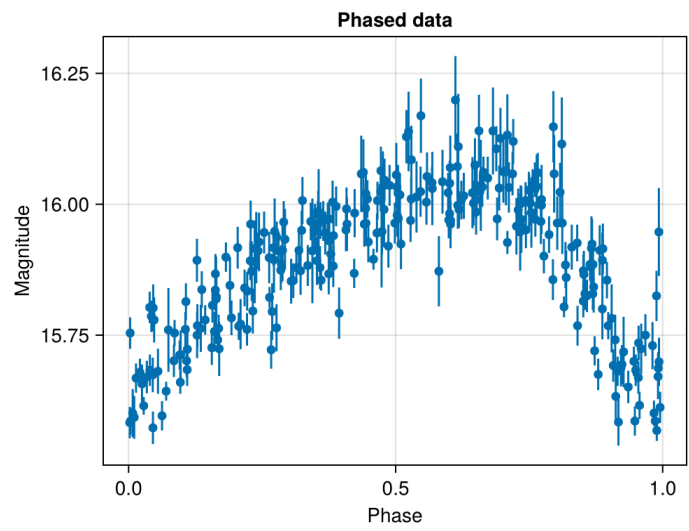
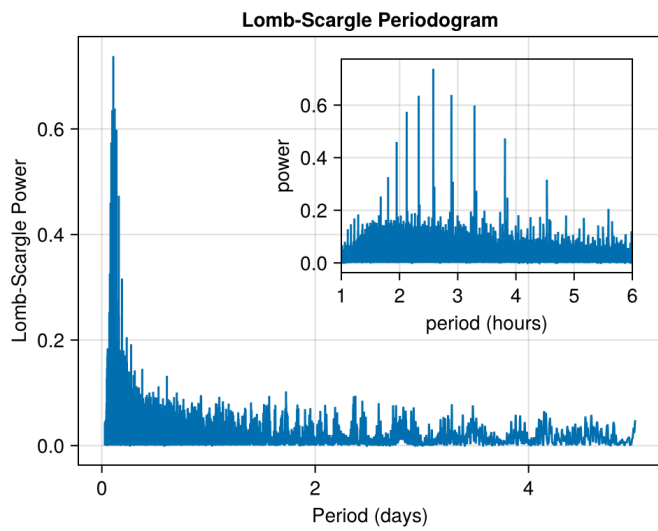


- The light-curve shows a very irregular monitoring, likely due to seasonal visibility of the source

Number of analyzed periods: 348039

Best period: **2.58** hours

- Once identified a period, P , computing the phase, ϕ , for each input observation time index is simple: $\phi_j = t_j/P - \text{int}(t_j/P)$.



Try to phase with a wrong period (~30% shorter)? ☐

- The identified period at ~ 2.58 hours looks fine and the phased light curve seems consistent with the noise of the data.
- Let's now move to data obtained by a survey carried out with a orbital instrument.

Let's study a variable source from the space Kepler mission:
KIC7198959

	TIME	TIMECORR	CADENCENO	SAP_FLUX	SAP_FLUX_ERR	SAP_BKG	SAP_BKG_ERR	F
1	169.52	0.00276929	2965	1.04089f7	80.6514	17445.8	6.81535	N
2	169.541	0.00276986	2966	1.02251f7	79.9479	17458.6	6.81904	N
3	169.561	0.00277044	2967	1.00993f7	79.4513	17447.1	6.81976	N
4	169.582	0.00277102	2968	1.01336f7	79.5974	17440.8	6.83005	N
5	169.602	0.0027716	2969	1.02778f7	80.1858	17444.1	6.82776	N

- Among the many columns we are here interested in the "TIME", "SAPFLUX" and "SAPFLUX_ERR" columns.

	TIME	SAP_FLUX	SAP_FLUX_ERR
1	169.52	1.04089f7	80.6514
2	169.541	1.02251f7	79.9479
3	169.561	1.00993f7	79.4513
4	169.582	1.01336f7	79.5974
5	169.602	1.02778f7	80.1858

- FITS files usually carry headers full of important information.

```

headhd = XTENSION= 'BINTABLE'           / marks the beginning of a new HDU
        BITPIX   =          8           / array data type
        NAXIS    =          2           / number of array dimensions
        NAXIS1   =        100           / length of first array dimension
        NAXIS2   =       4354           / length of second array dimension
        PCOUNT   =          0           / group parameter count (not used)
        GCOUNT   =          1           / group count (not used)
        TFIELDS  =         20           / number of table fields
        TTYPE1   = 'TIME'              / column title: data time stamps
        TFORM1   = 'D'                  / column format: 64-bit floating point
        TUNIT1   = 'BJD - 2454833'      / column units: barycenter corrected JD
        TDISP1   = 'D14.7'             / column display format
        TTYPE2   = 'TIMECORR'           / column title: barycenter - timeslice correction
        TFORM2   = 'E'                  / column format: 32-bit floating point
        TUNIT2   = 'd'                  / column units: day
        TDISP2   = 'E13.6'             / column display format
        TTYPE3   = 'CADENCENO'           / column title: unique cadence number
        TFORM3   = 'J'                  / column format: signed 32-bit integer
        TDISP3   = 'I10'                / column display format
        TTYPE4   = 'SAP_FLUX'           / column title: aperture photometry flux
        TFORM4   = 'E'                  / column format: 32-bit floating point
        TUNIT4   = 'e-/s'               / column units: electrons per second
        TDISP4   = 'E14.7'             / column display format
        TTYPE5   = 'SAP_FLUX_ERR'       / column title: aperture phot. flux error
        TFORM5   = 'E'                  / column format: 32-bit floating point
        TUNIT5   = 'e-/s'               / column units: electrons per second (1-sigma)
        TDISP5   = 'E14.7'             / column display format
        TTYPE6   = 'SAP_BKG'            / column title: aperture phot. background flux
        TFORM6   = 'E'                  / column format: 32-bit floating point
        TUNIT6   = 'e-/s'               / column units: electrons per second
        TDISP6   = 'E14.7'             / column display format
        TTYPE7   = 'SAP_BKG_ERR'       / column title: ap. phot. background flux error
        TFORM7   = 'E'                  / column format: 32-bit floating point

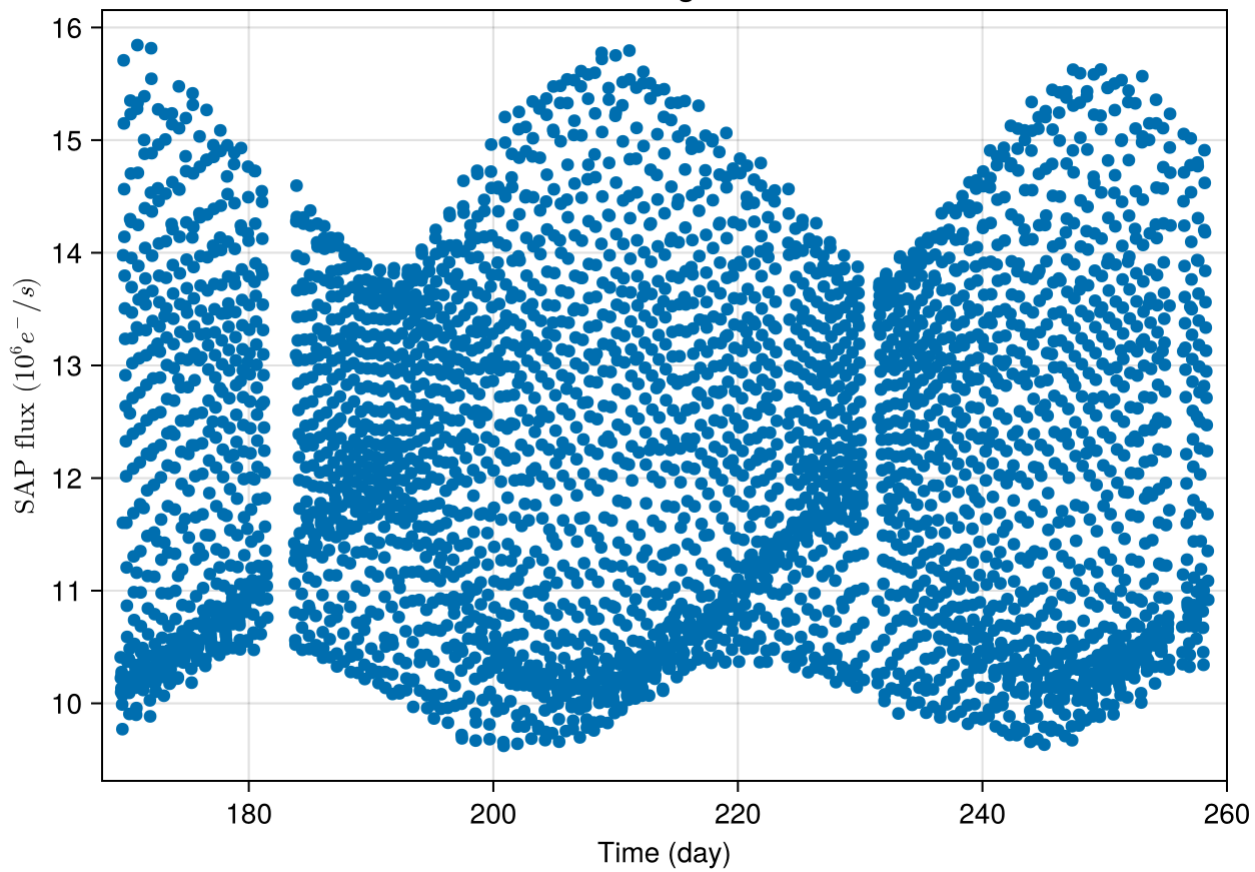
```

- Let's remove "NaN" entries from our dataset.

	TIME	SAP_FLUX	SAP_FLUX_ERR
1	169.52	1.04089f7	80.6514
2	169.541	1.02251f7	79.9479
3	169.561	1.00993f7	79.4513
4	169.582	1.01336f7	79.5974
5	169.602	1.02778f7	80.1858

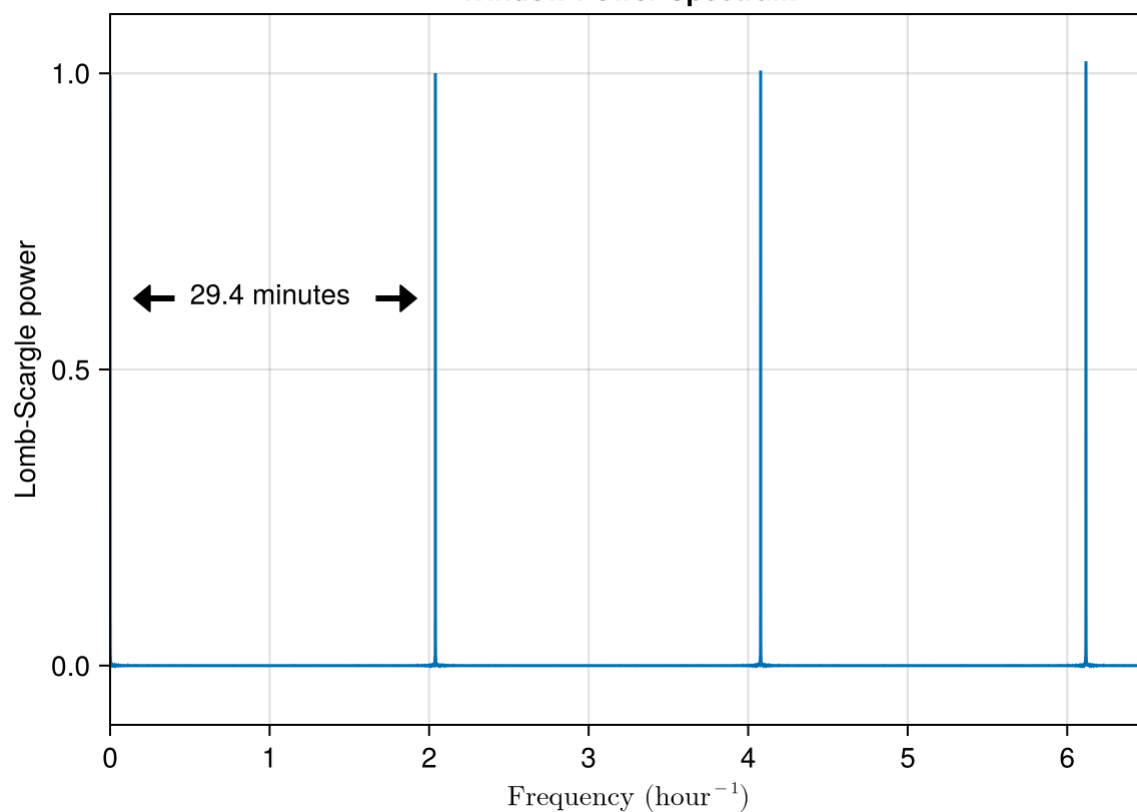
- Once we have a cleaned time-series let's plot it.

Observed light curve

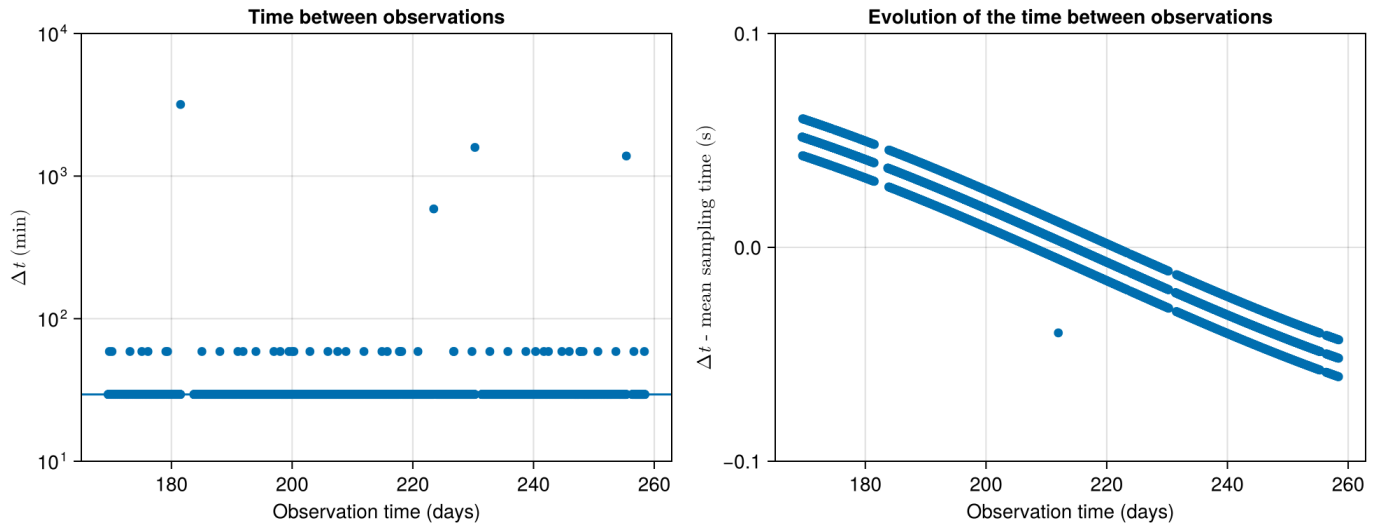


- Undoubtedly, this plot looks somehow weird. We anyway see there are gaps and a global modulation of brightness of the source.
- Let's now study the observing window.

Window Power Spectrum

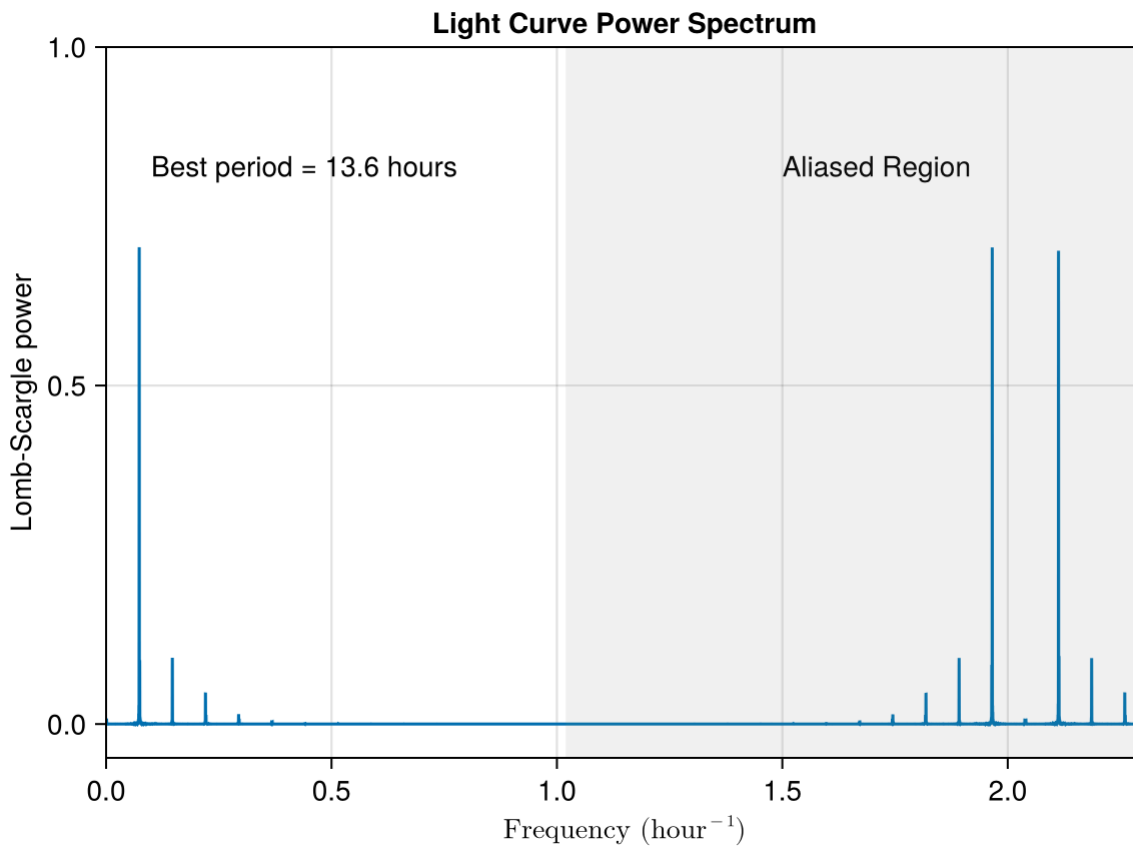


- We see a regular grid spaced by about 29.4 minutes at the beginning of the monitoring. However, even independently of the gaps, it is not an even sampling as shown below:



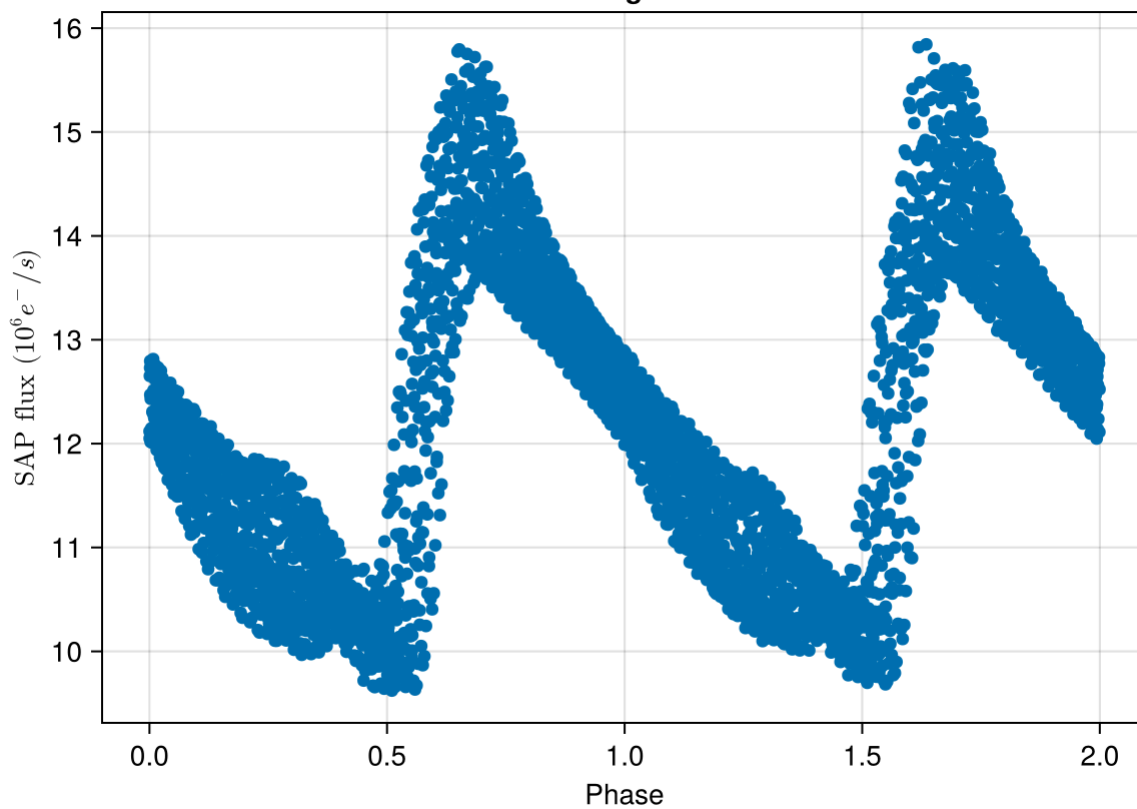
Mean sampling time: **29.42** minutes

- There are gaps, shorter interruptions and a drift in interval between an observation and the next.
- Let's now compute the LS periodogram for the light-curve.



- There is a well defined maximum and, due to the "quasi-regular" sampling even an aliased region.
- Finally, let's check whether the phased curve is reasonable.

Phased Light Curve



- The obtained light curve confirms the classification of this object as a RR-Lyrae variable.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Palaversa et al. \(2013\) - "Exploring the Variable Sky with LINEAR. III. Classification of Periodic Light Curves"](#)

Further Material

Papers for examining more closely some of the discussed topics.

- [GCVS - General Catalog of Variable Stars](#)

Credits

This notebook contains material obtained from

https://github.com/jakevdp/PracticalLombScargle/blob/master/figures/LINEAR_Example.ipynb and

<https://github.com/jakevdp/PracticalLombScargle/blob/master/figures/Kepler.ipynb>.

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about irregular sampling	Lecture about wavelet analysis	Course Summary
html	Lecture about irregular sampling	Lecture about wavelet analysis	Course Summary

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